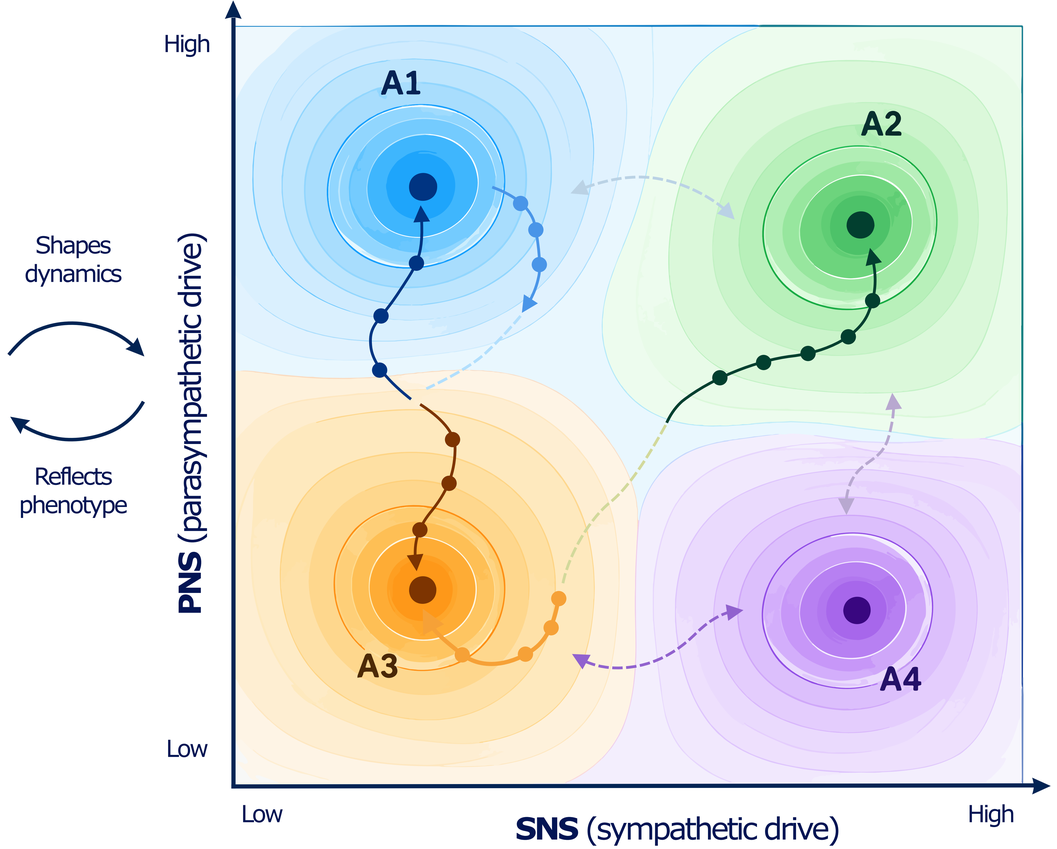
Multidomain Phenotype
(vector of signatures)

- **Brain / Cognitive** (EEG, cognitition, mood)
- **Molecular** (BDNF, inflammation, etc.)
- **Behavioral** (activity, sleep, nutrition)
- **Social** (support, context)
- **Autonomic / Cardiac** (HRV, RR dynamics)
- **Others** (metabolic, etc.)



Shapes dynamics

Reflects phenotype

Attractors
Stable states of the system

Trajectories
Individual dynamics over time

Transitions
Movement between atractors

Basins of attraction
Regions of initial conditions that converge to the same attractor

Associated with

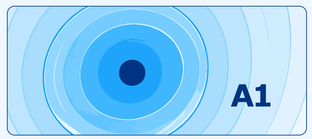
Predisposes to

Clinical Outcomes

- **Cardiovascular** (CVD, mortality)
- **Cognitive** (dementia, frailty)
- **Inflammatory / Immune** (chronic disease)
- **Functional** (physical, mental)
- **Others** (e.g., metabolic)

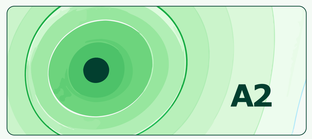
Examples: Phenotype → Topology → Outcome

Phenotype A (adaptive)
- High cognitive function
- High PNS tone
- Low inflammation
- Good sleep



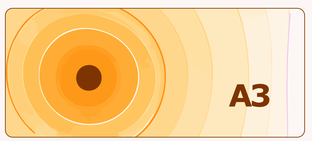
Favorable Outcomes

Phenotype B (resilient)
- Balanced autonomic tone
- High social support
- Healthy behavior
- Low stress



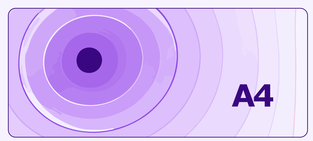
Better Prognosis

Phenotype C (vulnerable)
- Low SNS tone
- Low PNS tone
- Elevated inflammation
- Poor sleep



Higher Risk

Phenotype D (dysregulated)
- Variable dynamics
- Frequent transitions
- Lower stability
- Cognitive decline



Adverse Outcomes

Exposome ⟷ Multidomain Phenotype ⟷ Autonomic Cardiac Topology ⟷ Clinical Outcomes

Dynamic, bidirectional, context-dependent